

Sachin Bulchandani

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[@LinkedIn](#) | [@Portfolio](#) | Permanent Resident of Canada

PROFESSIONAL SUMMARY

Results-driven Cloud and DevOps Engineer with 3+ years of experience designing, automating, and optimizing cloud infrastructure across AWS, Azure, and GCP. Strong background in building CI/CD pipelines, containerized environments, and infrastructure-as-code solutions using Terraform and Kubernetes. Experienced in implementing ML-based systems for anomaly detection and data automation, improving operational efficiency and system reliability. Committed to delivering secure, scalable, and high-performing cloud solutions in Agile, collaborative environments.

SKILLS

Project Methodologies: Agile (Scrum), Waterfall, Kanban, SDLC

Cloud Platforms: AWS (EKS, EC2, S3, CloudFront, ALB, WAF, Route 53, Cognito, DynamoDB, RDS/Aurora, ElastiCache/Redis, OpenSearch, ECR, Lambda, Glue, Athena, SageMaker, CloudWatch, IAM, Secrets Manager, ACM, VPC, SNS, SQS, Event Bridge), Azure (VMs, Functions, Blob Storage, Databricks), GCP (BigQuery, AI Platform, Compute Engine, Cloud Run, Artifact Registry), IBM Cloud

DevOps & Automation: CI/CD (GitHub Actions, AWS Code Pipeline, Code Build, Code Deploy), Docker, Kubernetes (EKS), Git, Apache Airflow, Linux (Ubuntu, Red Hat), Windows Server

Programming & Scripting: Python, Bash, PowerShell, SQL

Infrastructure as Code: AWS CloudFormation, Terraform, AWS CDK (Python)

Data Technologies: ETL pipelines, Amazon Redshift, Apache Spark, SQL & NoSQL databases (DynamoDB, Aurora PostgreSQL, Redis)

WORK EXPERIENCE

Cloud DevOps Engineer

Kinaxis Inc., ON, Canada

Nov 2023 – Present

- Designed and managed scalable AWS infrastructure using EKS, EC2, RDS, and CloudFront, ensuring reliable multi-region deployments and improving system uptime across environments.
- Built and streamlined CI/CD pipelines with GitHub Actions and AWS Code Pipeline, reducing deployment time by 40% and minimizing rollback incidents during production releases.
- Containerized and orchestrated microservices using Docker and Kubernetes (EKS) to enable faster scaling, lower compute costs, and more consistent application delivery.
- Automated provisioning and configuration through Terraform and CloudFormation, which standardized infrastructure builds and eliminated frequent environment drift issues.
- Implemented secure access controls using IAM, Secrets Manager, and WAF, reinforcing compliance and reducing vulnerabilities identified during internal audits.
- Developed automation scripts in Python and Bash for monitoring, cost tracking, and recovery tasks, cutting manual intervention during outages and optimizing cloud spend.
- Partnered with development and QA teams under Agile sprints to refine deployment strategies, improving release predictability and fostering cross-team accountability.

Fraunhofer IOSB-INA — (Part-time – Remote)

Apr 2021 – Oct 2022

Research Assistant NRW, Germany

- Developed and fine-tuned an anomaly detection pipeline using LSTM-VAE and ensemble learning, reducing false positives in cyber-attack detection by 10% through better temporal feature extraction.
- Applied synthetic oversampling techniques (SMOTE) to rebalance skewed attack datasets, boosting model recall and improving sensitivity to rare but critical threat patterns.
- Mapped and validated model outputs against the MITRE ATT&CK framework to benchmark coverage, closing detection gaps in 15% of evaluated adversarial techniques.
- Processed and transformed multi-stream sensor data using Python, NumPy, and Pandas, engineering time-series features that enhanced anomaly classification accuracy.
- Authored and presented research on deep-learning-based ICS cybersecurity Modeling, communicating technical findings and model performance improvements to senior engineers and academic reviewers.

PROJECTS

Scalable E-Commerce Platform on AWS Personal Graduation Project | [@E-Commerce Project](#)

- Architected and deployed a multi-AZ, fault-tolerant AWS infrastructure, achieving near-zero downtime for critical workloads.
- Designed and containerized a secure microservices backend using Docker, orchestrating services like Product, Order, and User/Auth on Amazon EKS for scalability.
- Strengthened security posture with AWS WAF, IAM role-based access, and network segmentation via VPCs and Security Groups.
- Engineered an event-driven notification system for real-time order processing and user communication using AWS SNS and SQS.
- Leveraged DynamoDB, Aurora PostgreSQL, and ElastiCache for Redis to optimize data storage and query performance.
- Automated deployments through a CI/CD pipeline (GitHub Actions + AWS Code Pipeline), cutting release time from hours to minutes.

Movielytics: Movie Data Warehouse & ETL Pipeline Personal Project | [@Movielytics ETL Project](#)

- Designed and implemented an AWS-based ETL pipeline processing 45K+ records with PySpark and loading into a Redshift data warehouse.
- Orchestrated complex ETL workflows using Apache Airflow, with all components containerized in Docker for portability.
- Embedded custom data quality checks within Airflow DAGs to ensure high-integrity datasets.
- Modeled the warehouse using a star schema in Amazon Redshift, reducing query execution times by 40%.
- Enabled analytics to uncover trends in user engagement and revenue drivers for data-driven decision-making.

Breast Cancer Detection (Computer Vision) Personal Project | [@Breast Cancer Classification \(Live Model\)](#)

- Built and deployed a computer vision application for medical image analysis, hosted as a public service on Google Cloud Run.
- Fine-tuned a RESNET50V2 model achieving 93% classification accuracy on the BreakHis v1 dataset for binary breast cancer detection.
- Established a reproducible ML workflow using DVC for data versioning and CI/CD for automated deployments.
- Developed an ETL pipeline to clean and partition histopathological images, ensuring balanced and unbiased training.

Intelligent Job Scheduling with GNN-RL (Master's Thesis, Fraunhofer IOSB-INA) | [@Master Thesis](#)

- Designed a GNN-based RL agent for Job-Shop Scheduling with a custom Python simulation on AWS EC2.
- Achieved an 18% improvement in average job completion time compared to traditional heuristics.

RL for FMS Routing (Research Project, Fraunhofer IOSB-INA) | [@Research Project](#)

- Implemented a PPO-based RL algorithm in PyTorch to optimize material routing in a Petri-Net simulated FMS.
- Reduced routing collisions by 30% while improving throughput in dynamic production scenarios.

EDUCATION

Post Graduate Certificate in Cloud Computing Technologies | George Brown College, Toronto, ON

MSc. in Information Technology | Technical University of Ostwestfalen-Lippe, NRW, Germany

B-Tech in Electrical Engineering | Charotar University of Science and Technology, Gujarat, India

CERTIFICATIONS

1. AWS Certified Solutions Architect – Associate (Expected Dec 2025)
2. AWS Cloud Practitioner – Amazon Web Services
3. AWS Certified ML Specialist – Amazon Web Services
4. Advanced Machine Learning – IBM on Coursera
5. Professional Data Analytic – Google on Coursera